

# User Study – ML2++ (Textual DSL)

*Instructions:* Fill Section A once. For Section B, answer for *each* use case you worked on. If you did multiple use cases, **duplicate Section B** per use case.

**Participant ID:** \_\_\_\_\_ **Use case(s):** ☒ U1 River Flow ☐ U2 Smart-Home Energy  
☐ U3 Solar Energy

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## Section A – Background (answer once)

*This section helps us understand your experience level so we can analyze results fairly.*

### A1. Educational level (highest completed)

*Tell us the highest level of education you have finished.*

☐ High School ☐ Vocational/Technical ☐ Bachelor's ☒ Master's ☐ Doctorate ☐ Other: \_\_\_\_\_

### A2. Primary role (pick the closest)

*Choose the option that best matches your current job or main activity. If none fits, select “Other”.*

☐ IoT/CPS Developer ☐ MDE Practitioner ☐ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☒ Software Architect / Full-stack ☐ System Integrator ☐ Early Adopter / Innovator ☐ Other: Devops

### A3. Self-rated expertise (tick one per row)

*For each area, select your skill level. “IoT/CPS” = connected devices/systems. “MDE” = using models to guide or generate code. “ML/Data Science” = working with data and ML.*

|                                 | Beginner                            | Intermediate                        | Advanced                 | Expert                   |
|---------------------------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|
| IoT/CPS development             | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |
| Model-Driven Engineering (MDE)  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |
| Machine Learning / Data Science | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

### A4. Prior experience (used DSLs/modeling tools/ML frameworks before?)

*Have you used similar tools (e.g., modeling editors, DSLs, ML libraries/frameworks)?* tensorflow

☒ Yes ☐ No

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## Section B – Task-Specific (duplicate this page *per use case*)

**This B-section is for:** ☒ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

*If you did more than one use case, copy this whole Section B for each one and fill it separately.*

## B1. Time & Effort

Please record your times honestly. If unsure, give your best estimate. “Extra learning time” = time reading docs/tutorials to figure things out.

### B1. Time spent (minutes)

Minutes you spent defining/editing the DSL, generating code, reviewing outputs, and any extra learning time.

| ML2++ (Textual DSL)              |              |
|----------------------------------|--------------|
| Defining DSL + generate + review | <u>60min</u> |
| Extra learning time              | <u>45min</u> |

### B2. Statements written (DSL) (exclude blanks/comments)

Approximate number of DSL lines/statements you actually typed or edited.

| ML2++ (Textual DSL)  |            |
|----------------------|------------|
| ML2++ DSL statements | <u>156</u> |

## B2. Task Completion & Ease of Use

Tell us if you finished the task and how easy/hard each step felt. Then write how many minutes you spent per step.

### B3. Were you able to finish the use case?

“None” = could not complete; “Partial” = finished some parts; “Complete” = finished everything.

☐ None   ☐ Partial   ☒ Complete

### B4. Ease by workflow step (with per-step time)

Rate difficulty (1=very easy, 5=very difficult) and write minutes spent. “Model configuration” = choosing algorithm and hyperparameters. “Model evaluation” = checking results with plots/metrics.

| Step                                                    | 1                                   | 2                                   | 3                                   | 4                        | 5                        | Time            |
|---------------------------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|-----------------|
| Whole workflow (end-to-end)                             | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <u>        </u> |
| Data preprocessing                                      | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <u>        </u> |
| <b>Model configuration</b> (algorithm, hyperparameters) | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <u>        </u> |
| Model training                                          | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <u>        </u> |
| Model evaluation (metrics/plots/inspection)             | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <u>        </u> |

Tip: use a simple stopwatch and pause for breaks. If unsure, estimate.

### B5. Ease of ML2++ elements (1=Very easy → 5=Very difficult)

Rate how easy it was to work with each part of the language/editor.

| Element                                                        | 1                        | 2                                   | 3                        | 4                        |
|----------------------------------------------------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|
| Data types & objects                                           | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Things & behaviours (statecharts)                              | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Ports & properties                                             | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| <b>Model configuration options (algorithm/hyperparameters)</b> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Configuration (instances/connectors)                           | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

### B3. Learning & Usability

*These questions focus on how quickly you learned the tool and how natural it felt.*

#### B6. Learning curve (1=Very easy → 5=Very difficult)

*How much effort did it take to learn enough to finish this use case?*

☐1 ☐2 ☒3 ☐4 ☐5

#### B7. Intuitiveness (1=Very intuitive → 5=Not intuitive)

*Did the tool feel natural/obvious to use, or did it feel confusing?*

☐1 ☐2 ☒3 ☐4 ☐5

#### B8. Hardest part to learn (briefly)

*Name the single most confusing or time-consuming thing to learn.*

learn how it work

### B4. Debugging & Error Handling

*Tell us about problems you faced and how hard they were to fix.*

#### B9. Error frequency (0=None, 1=Few, 2=Moderate, 3=Many)

*How many problems blocked you (errors, failed runs, missing packages, wrong paths, etc.)?*

☐0 ☐1 ☒2 ☐3

#### B10. Clarity of error messages (1=Not clear → 5=Very clear)

*When something went wrong, did the messages help you understand the fix?*

☐1 ☒2 ☐3 ☐4 ☐5

#### B11. Debugging effort (1=Easy → 5=Hard)

*Overall, how hard was it to fix your issues?*

☐1 ☒2 ☐3 ☐4 ☐5

#### B12. Debugging time (minutes) (estimate)

*About how many minutes did you spend fixing problems (total for this use case)?*

20 min

## B5. Code Quality, Maintainability, Flexibility

These questions refer to the generated code you viewed and how easy it would be to change things later.

### B13. Generated code quality (1=Poor $\rightarrow$ 5=Excellent)

How good is the generated code (readable, well-structured, correct)?

☐1 ☐2 ☐3 ☒4 ☐5

### B14. Maintainability (1=Very difficult $\rightarrow$ 5=Very easy)

If you had to update/change your solution later (e.g., new dataset, new horizon), how easy would it be?

☐1 ☐2 ☒3 ☐4 ☐5

### B15. Flexibility / Customisation (1=None $\rightarrow$ 5=Very flexible)

How easily can you change models, hyperparameters, thresholds, or add steps without workarounds?

☐1 ☐2 ☐3 ☒4 ☐5

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## Section C – Overall Assessment (ML2++ only)

Your overall opinion and which features helped you most.

### C1. Overall satisfaction (1=Very dissatisfied $\rightarrow$ 5=Very satisfied)

Your overall feeling about using ML2++ for this use case.

☐1 ☐2 ☐3 ☐4 ☒5

### C2. Would you use ML2++ for your next similar project?

If you had a similar project tomorrow, would you choose this approach? Tell us briefly why.

☒ Yes ☐ No ☐ Undecided Why? because you can focus on your problem and design our model better

### C3. Feature effectiveness (1=Ineffective $\rightarrow$ 5=Very effective)

How useful were these features for you?

| Feature                         | 1                        | 2                        | 3                                   | 4                                   | 5                        |
|---------------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
| Time-series forecasting models  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Automatic preprocessing & plots | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |
| Training configuration options  | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |
| Multi-day forecasting accuracy  | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |

### C4. Favourite feature: define our models

What was the single most helpful feature?

**C5. Least favourite feature:** preprocessing

*Which feature felt least helpful or got in your way?*

**C6. Missing features you would like:** design to have better syntax i think yaml file its good

*What do you wish the tool had (e.g., a plot, algorithm, wizard, or shortcut)?*

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## Section D – Open Feedback

*Tell us anything else that would help us improve.*

**D1. Barriers to adoption** (*what would stop you from using ML2++ in production?*)

*Examples: setup, missing plots, docs, integration, performance, governance.*

low usage on community and lack of documentaction

**D2. One improvement you would prioritize**

*If we could fix/add only one thing, what would help you the most?*

support modular

use configuration file like yaml

try lower conf code

*Scale anchors used throughout for consistency:*

- Ease / Learning: 1=Very easy, 2=Easy, 3=Moderate, 4=Difficult, 5=Very difficult.
- Intuitiveness: 1=Very intuitive, 2=Intuitive, 3=Neutral, 4=Somewhat confusing, 5=Not intuitive.
- Error clarity: 1=Not clear, 2=Somewhat clear, 3=Clear, 4=Very clear, 5=Extremely clear.
- Code quality: 1=Poor, 2=Fair, 3=Good, 4=Very good, 5=Excellent.
- Maintainability: 1=Very difficult, 2=Difficult, 3=Moderate, 4=Easy, 5=Very easy.
- Flexibility: 1=None, 2=Limited, 3=Some, 4=Flexible, 5=Very flexible.
- Feature effectiveness: 1=Ineffective, 2=Low, 3=Moderate, 4=Effective, 5=Very effective.

# User Study – Manual Python Workflow (Method 1 Only)

*Instructions:* Fill Section A once. For Section B, answer *per use-case*. Duplicate Section B if you completed multiple use-cases.

Participant ID: \_\_\_\_\_ Your assignment: ☐ Method 1 (Manual Python)

Use case(s) you worked on: ☒ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

## Section A – Background (answer once)

### A1. Educational level (*highest completed*)

☐ High School ☐ Vocational/Technical ☐ Bachelor's ☒ Master's ☐ Doctorate ☐ Other: \_\_\_\_\_

### A2. Primary role (*pick the closest*)

☐ IoT/CPS Developer ☐ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☒ Software Architect / Full-stack ☐ System Integrator ☒ Other: \_\_\_\_\_

Devops

### A3. Self-rated expertise (*tick one per row*)

|                                  | Beginner                 | Intermediate                        | Advanced                            | Expert                   |
|----------------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
| Python programming               | <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/> |
| Time-series / ML                 | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |
| Domain knowledge (your use-case) | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> |

### A4. Prior experience (*used ML frameworks/libraries before?*) ☒ Yes ☐ No

## Section B – Task-Specific (duplicate this page *per use-case*)

This B-section is for: ☒ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

### B1. Time & Effort

*Record your own times. Estimate if needed. “Extra learning time” = time spent reading docs/tutorials.*

#### B1. Time spent (minutes)

|                                   | Method 1 (Manual Python) |
|-----------------------------------|--------------------------|
| Writing/testing code (end-to-end) | 14                       |
| Extra learning time               | 6                        |

### B2. Lines of code (LoC) (*exclude blanks/comments*)

Manual Python LoC: 108

## B2. Task Completion & Ease of Use

Completion: None = could not finish; Partial = finished some parts; Complete = all parts.

### B3. Were you able to finish the use-case with Manual Python?

Method 1 (Manual Python): ☐ None ☐ Partial ☒ Complete

### B4. Ease by workflow step (Manual Python) (1=Very easy, 5=Very difficult; tick one per cell)

| Step                               | 1                                   | 2                                   | 3                                   | 4                        | 5                        |
|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|
| Whole workflow                     | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |
| Data preprocessing                 | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Feature engineering (lags/rolling) | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |
| Model training                     | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |
| Model evaluation/plots             | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |

### B1. Which steps did you complete? (check all that apply)

- ☒ Preprocessing
- ☒ Model configuration
- ☒ Training
- ☒ Evaluation
- ☒ Plotting/Diagnostics

### B2. How difficult was each step? (1=Very easy, 5=Very difficult; tick one per row)

| Step                 | 1                                   | 2                        | 3                                   | 4                        | 5                        |
|----------------------|-------------------------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|
| Preprocessing        | <input type="checkbox"/>            | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Model configuration  | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |
| Training             | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |
| Evaluation           | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |
| Plotting/Diagnostics | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> |

### B3. Notes (issues, decisions, or gaps to revisit):

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### B5. Time spent per step (record for this use-case; minutes or start/end)

| Step                                 | Start (hh:mm) | End (hh:mm) | Minutes |
|--------------------------------------|---------------|-------------|---------|
| Preprocessing                        | 4             | 00          | 00      |
| Model configuration                  | 1             | 00          | 00      |
| Training                             | 1             | 00          | 00      |
| Evaluation (metrics/baseline)        | 1             | 30          | 00      |
| Plotting / Diagnostics               | 1             | 30          | 00      |
| Debugging (errors, crashes, outputs) | 5             | 00          | 00      |
| Extra learning / searching / docs    | 6             |             |         |
| Total                                |               |             | 20      |

*Notes:* “Learning” includes reading docs, web searches, tutorials, setup notes, and guides. “Debugging” covers error fixing, crashes, failed runs, and investigating unexpected results.

**B6. Comments (bugs, friction points, ideas):**

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LSTM train: 275s Test: 4s

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GRU train: 263s Test: 3s

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## Section E – IoT Use Case Template Reuse (Part 2)

**Instructions:** This section evaluates your experience reusing the provided IoT pipeline template (e.g., River Flow Forecasting) and adapting it for your own use case. Please answer honestly.

### E1. Template Understanding

How clear was the IoT pipeline template (overview, scripts, execution order)?

☐ 1 Very unclear ☒ 2 Unclear ☐ 3 Neutral ☐ 4 Clear ☐ 5 Very clear

### E2. Adaptation Effort

How easy was it to adapt the template to your chosen use case?

☐ 1 Very difficult ☒ 2 Difficult ☐ 3 Neutral ☐ 4 Easy ☐ 5 Very easy

### E3. Reuse of Offline Training Code

How straightforward was it to reuse the provided offline training code with your dataset?

☐ 1 Not at all ☐ 2 A little ☒ 3 Neutral ☐ 4 Mostly ☐ 5 Very straightforward

**E4. Time & Effort (minutes)** Understanding the template: \_\_\_\_\_  
Adapting scripts (sensors, gateway, analytics): 7 5 \_\_\_\_\_  
Running end-to-end pipeline: 60 \_\_\_\_\_

### E5. Problems & Debugging

Did you face errors while reusing the template? ☒ Yes ☐ No

If yes, briefly describe the main issues: \_\_\_\_\_ . package installing

How difficult were they to fix?

☐ 1 Very easy ☐ 2 Easy ☒ 3 Neutral ☐ 4 Hard ☐ 5 Very hard

### E6. Usefulness

How useful was the template for accelerating your implementation?

☐ 1 Not useful ☐ 2 Slightly useful ☐ 3 Neutral ☒ 4 Useful ☐ 5 Very useful

### E7. Suggested Improvements

What changes would make the IoT template easier to reuse for future use cases?

combine server and analytics and scale data i train give more data for training

*Scale anchors (for consistency):*

- Clarity / Ease: 1 = Very unclear/difficult, 2 = Unclear/difficult, 3 = Neutral, 4 = Clear/easy, 5 = Very clear/easy.
- Debugging difficulty: 1 = Very easy, 5 = Very hard.
- Usefulness: 1 = Not useful, 5 = Very useful.